You could argue that these scenarios are not exactly the same thing. The quantum fluctuation that re-creates your experience of stubbing your toe might produce something that is exactly like you in every detail, but you, as an entity, would have been long dead by then. But this brings up questions of what it means to be *you*. Is the exact configuration of atoms you, or is there something ineffable and persistent about your consciousness that could never be re-created piece by piece? This is the same question that sparks heated debates among science fiction fans around teleportation and whether or not Captain Kirk was brutally murdered every time he stepped into a transporter beam, only to be replaced by a duplicate impostor that mistakenly believed itself to be him. We are unlikely to answer it here.

But it does bring up another wrinkle to the rebirth-by- quantum-fluctuation scenario—one that has as much to do with the transporter question as it does with the sperm whale and bowl of petunias, all wrapped up in a kind of quantum mechanical solipsism. It's a problem called Boltzmann Brains.

The idea is that if it's possible for the entire universe to quantum-mechanically fluctuate out of the vacuum, it's much *more* likely for just a single galaxy to do it, because a single galaxy is less complicated and requires less stuff to suddenly appear. And if it's more likely for a single galaxy to appear, it's more likely for a single solar system, or a single planet. In fact, far more likely even than that is that the only thing that fluctuates out of the vacuum is a single human brain, one that contains all your memories and is in the process of imagining that it lives on a perfectly functional world and is currently sitting in a coffee shop typing the words to the fourth chapter of a book about the end of the cosmos.

The Boltzmann Brain problem is the assertion that this unfortunate brain, doomed to quantum-fluctuate back into the vacuum almost instantaneously after its creation, is so vastly more likely to occur than a whole universe that, if we want to use random fluctuations to build our universe, we have to accept that we're much more likely to be just imagining the whole thing.

This question is not yet settled. Despite being one of the first people to propose the Boltzmann Brain problem in this context, Albrecht now comes down on the side that it's more likely for a de Sitter universe to create a very low-entropy state like the Big Bang than something small on the brink of reabsorption. The basic argument is that creating a low-entropy state might seem to take a lot of quantum fluctuation energy, but actually takes